

UNIVERSITY OF OSLO

Analysis of the dark matter density profile of the globular cluster M15

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Master Thesis defense

Oslo, June 11th 2025



A. Block/Mt. Lemmon SkyCenter/Univ. of Arizona



Contents

- 1 Context
- 2 CLUMPY
- 3 Tidal radius impact
- 4 Testing different models
- 5 Model comparison
- 6 Conclusions



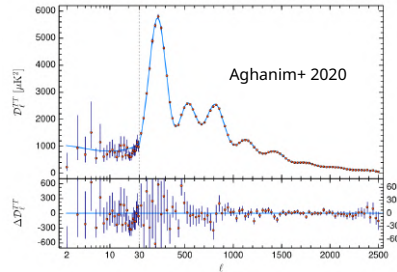
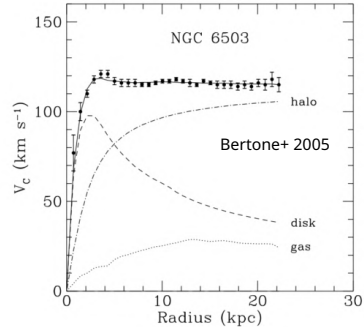
The quest for Dark Matter

- Dark Matter (DM):
 - 25% of the Universe's total energy content.
 - 85% of the Universe's matter.



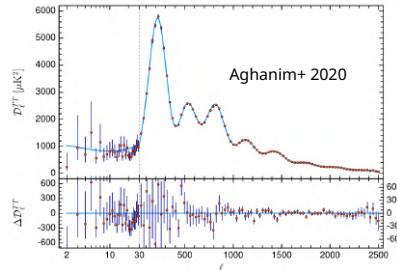
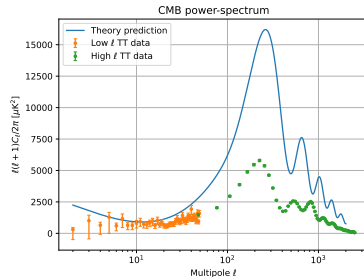
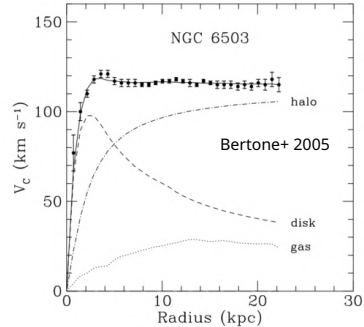
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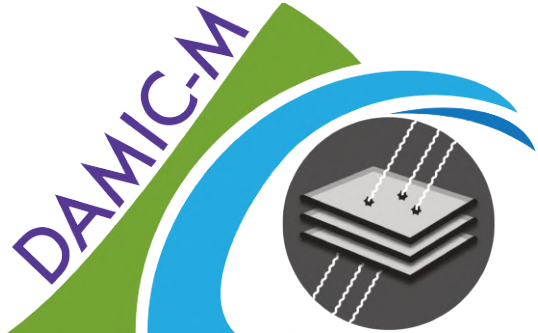
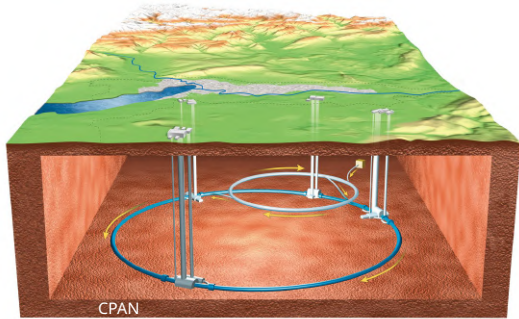


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DM searches



DAMIC-M collaboration

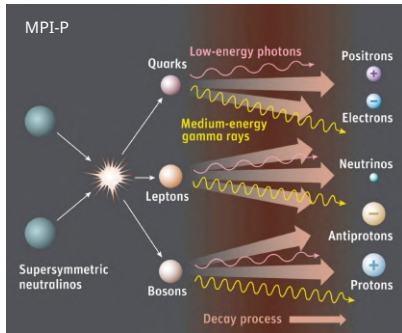
DM searches

- DM cross-section: extremely small $\Rightarrow$ very rare interaction.

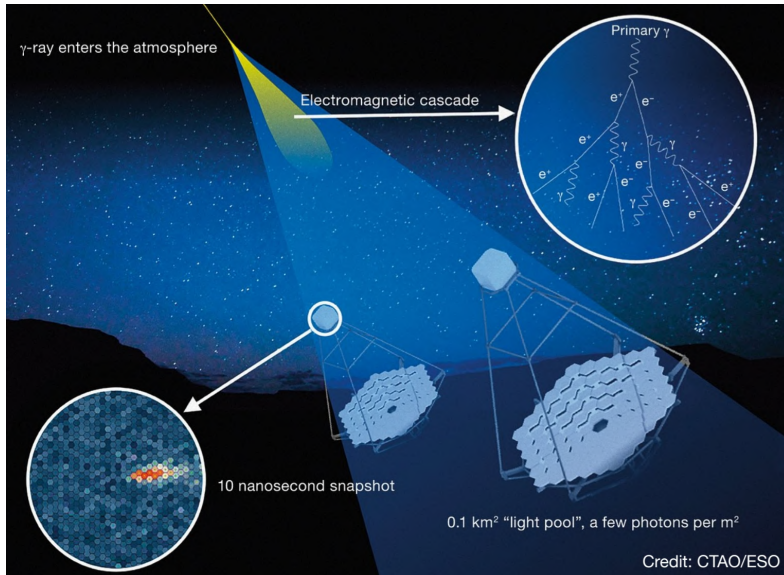


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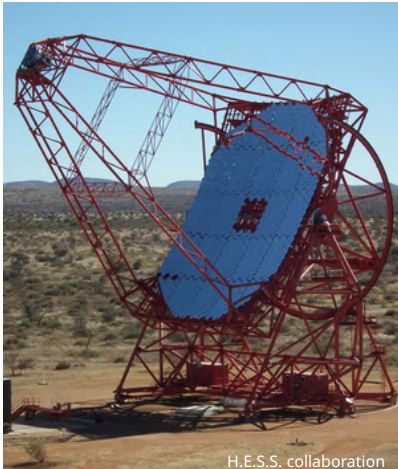
- DM cross-section: extremely small $\Rightarrow$ very rare interaction.
- Self-interacting DM particles (WIMPs) $\Rightarrow$ Standard Model (SM) particles: **very high energy (VHE) γ -rays.**



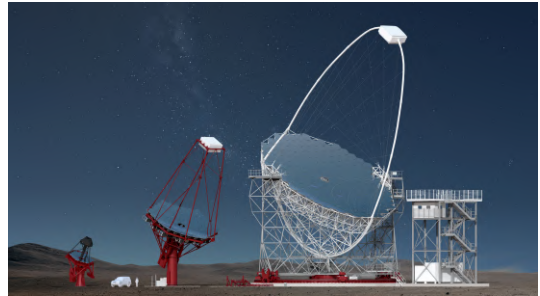
VHE γ -ray detection



DM indirect searches: IACTs



H.E.S.S. collaboration



Gabriel Pérez Díaz, IAC

MAGIC

- Two Imaging Atmospheric Cherenkov Telescopes located at the ORM in La Palma, Canary Islands (Spain).
- 17m diameter each (MAGIC-I 2004, MAGIC-II 2009).
- Angular resolution $\sim 0.1^\circ$.
- FoV $\sim 3.5^\circ$.
- Energy range from $\sim 20\text{GeV}$ up to $\sim 100\text{TeV}$.
- Energy resolution $\sim 15 - 25\%$.

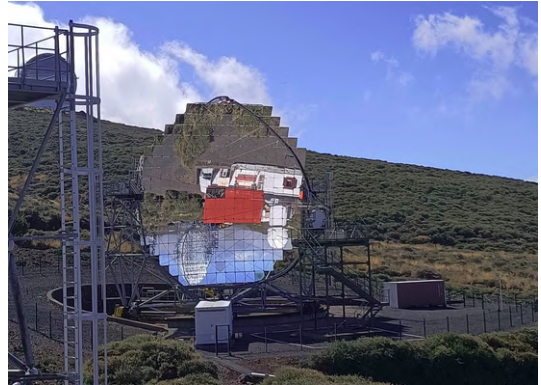


Image taken during my technical shift last semester.

DM indirect searches

$$\frac{d\Phi_{\text{ann}}}{dE_\gamma} = \frac{\langle \sigma_{\text{ann}} v \rangle}{8\pi m_\chi^2} \sum_i \text{BR}_i \frac{dN_\gamma^{(i)}}{dE_\gamma} \cdot J(\Delta\Omega)$$

DM indirect searches

Thermally average annihilation cross section

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Targets of interest

- dSphs / Globular clusters
- Galaxy clusters
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Targets of interest

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 - DM-dominated
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 - No background emission
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 - Relevant background contamination
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$$J_{\text{M15}} \sim J_{\text{MW}}$$

M15

- Solar distance: 10.9 kpc.
(Bhardwaj+ 2021).
- Galactocentric distance: 10.4 kpc.
(Harris 1996, Kayser+ 2008)
- Luminosity: $3.4 \cdot 10^5 L_{\odot}$.
- Mass: $5.6 \cdot 10^5 M_{\odot}$.
(Marks & Kroupa 2010)



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MAGIC observations

- Total effective exposure time: 136.5 hours.



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MAGIC observations

- Total effective exposure time: 136.5 hours.
- **Data analysis to be finalized.**



Daniel López / IAC



A. Block/Mt. Lemmon SkyCenter/Univ. of Arizona

Previous works

- DM amount in M15 already estimated by Whipple (Wood+ 2008) and H.E.S.S. (Abramowski+ 2011)

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- Here I recomputed J through the **MCMC Jeans analysis performed with CLUMPY**. (Hütten+ 2019)

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Inputs and parameters

- Brightness profile: King 2D (Newell & O'Neil 1978)
 - Scale density: $\Sigma_s^* = 7.3 \cdot 10^{10} L_{\odot} / \text{kpc}^2$.
 - Scale radius: $R_s^* = 5.5 \cdot 10^{-4} \text{ kpc}$.

$$\Sigma^*(r) = \Sigma_s^* \left[\frac{1}{\sqrt{1+(r/R_s^*)^2}} - \frac{1}{\sqrt{1+(R_t/R_s^*)^2}} \right]^2 \quad \text{CLUMPY}$$

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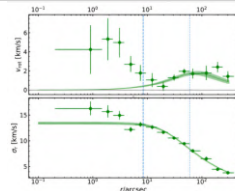
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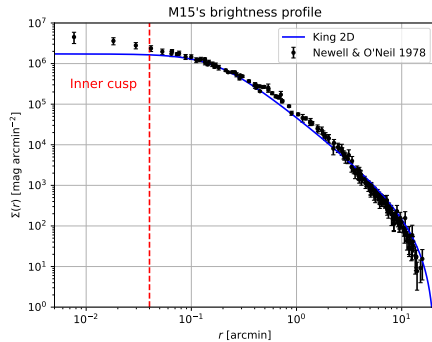
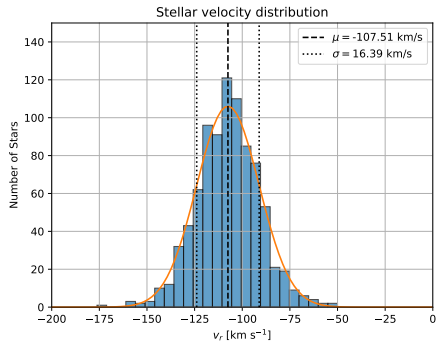
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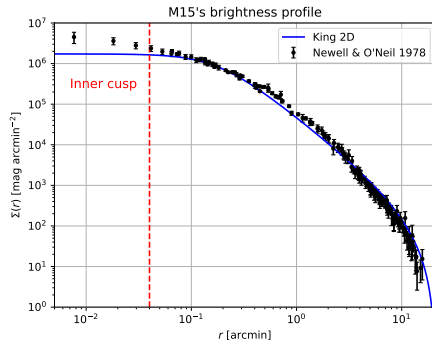
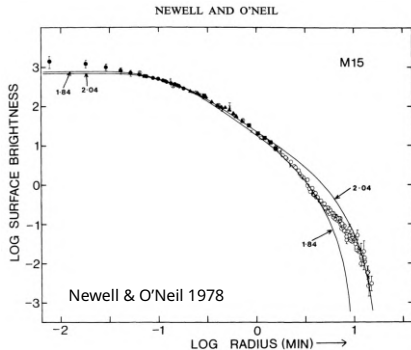


Inputs and parameters



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$$C = \log_{10} \left(\frac{R_t}{r_c} \right) \text{ with } r_c = 7.3 \cdot 10^{-4} \text{ kpc. (Peterson \& King 1975)}$$



2nd order Jeans equation

$$\frac{1}{n_*} \left[\frac{d}{dr} \left(n_* \bar{v}_r^2 \right) \right] + \frac{2}{r} \beta_{\text{ani}}(r) \bar{v}_r^2 = - \frac{G [M_{\text{DM}}(r) + M_*(r)]}{r^2}$$

M15's size

- Peterson & King (1975) $\implies R_t = 67$ pc.
- Newell & O'Neil (1978) $\implies R_t \in [58, 80]$ pc.
- This work $\implies R_t = 44$ pc.

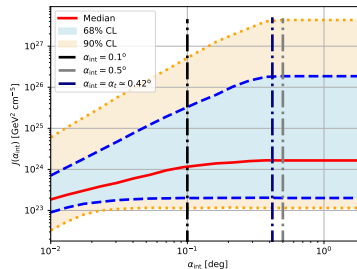
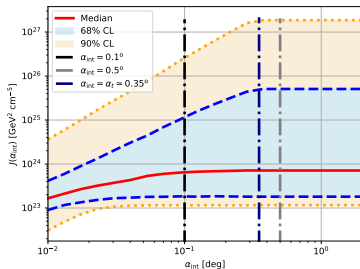
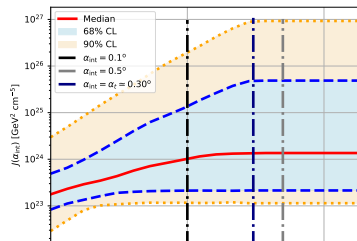
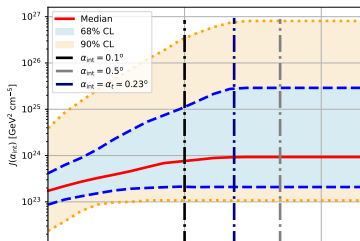


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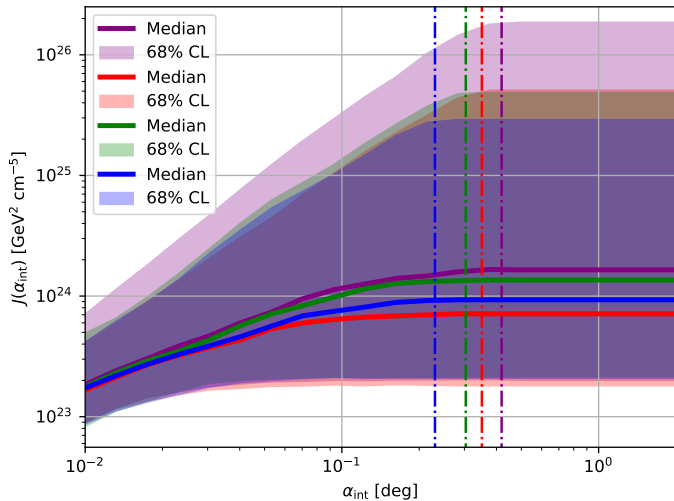
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Does it have an impact?



Tidal radius impact



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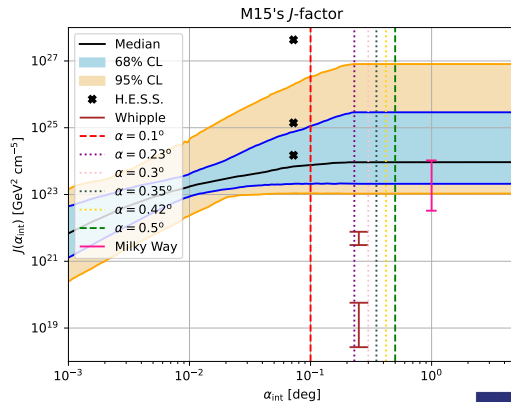
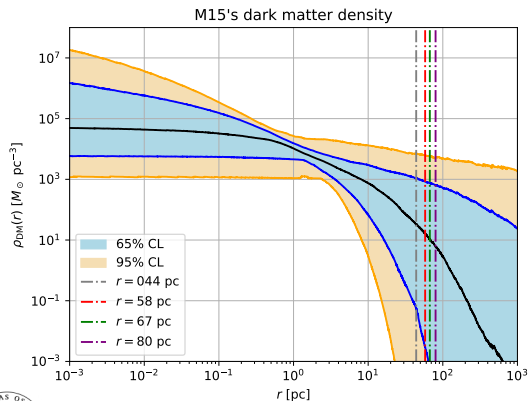
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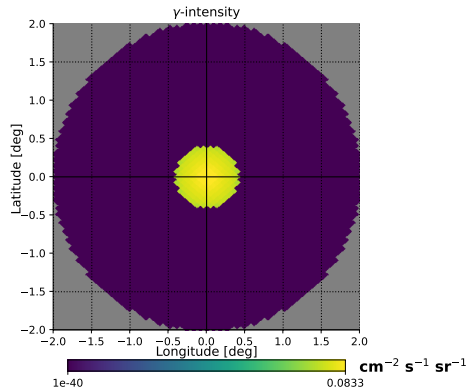
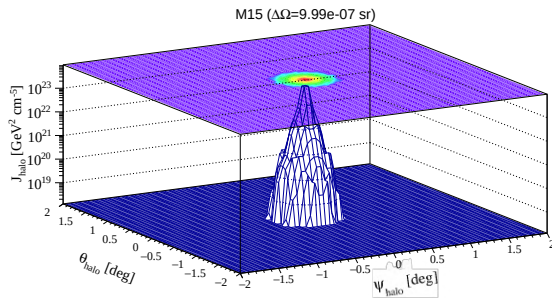
Einasto model (Einasto, 1965)

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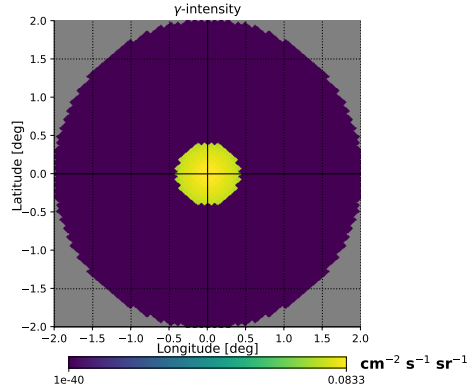
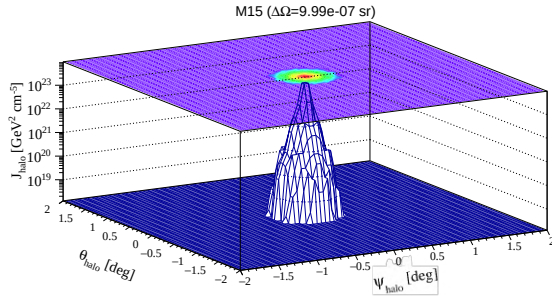
Results



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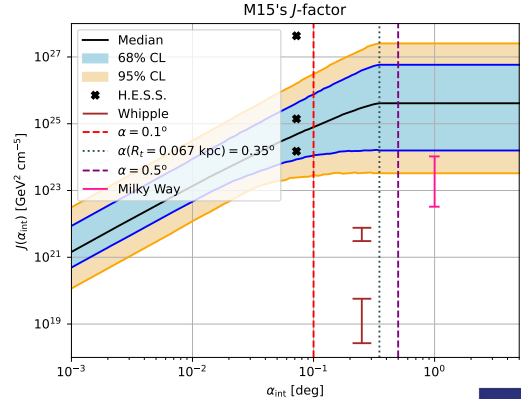
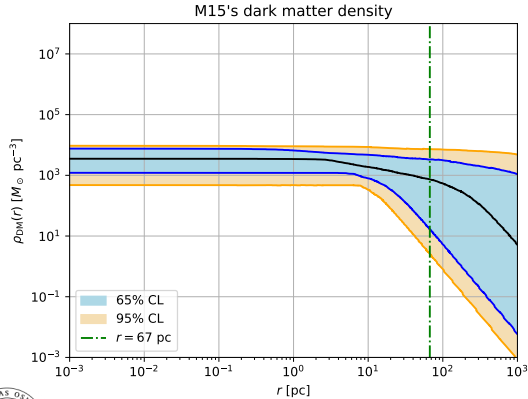


95% of the signal lies in a circle of radius $\sim 0.15^\circ \gtrsim$ MAGIC's resolution

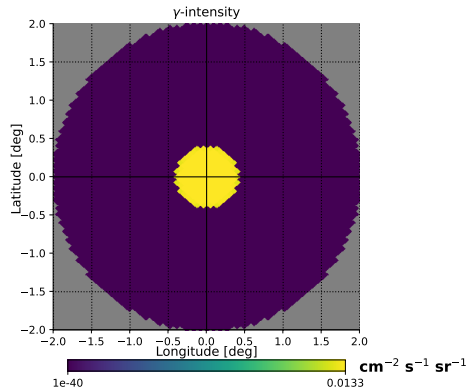
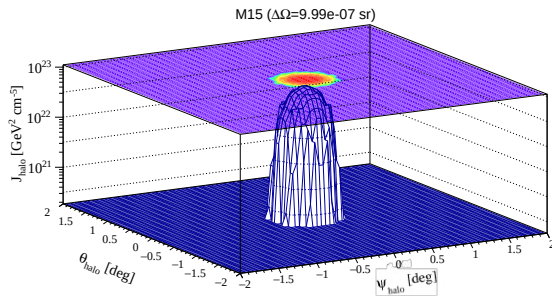
Burkert model (Burkert, 1995)

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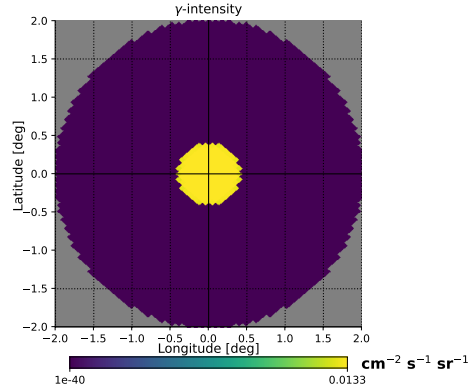
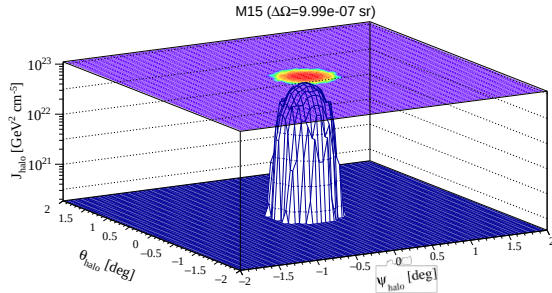
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Results



Results

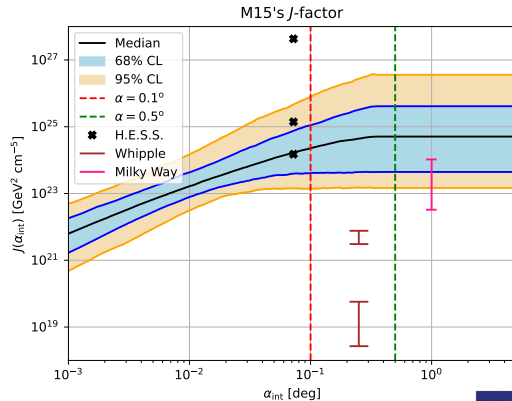
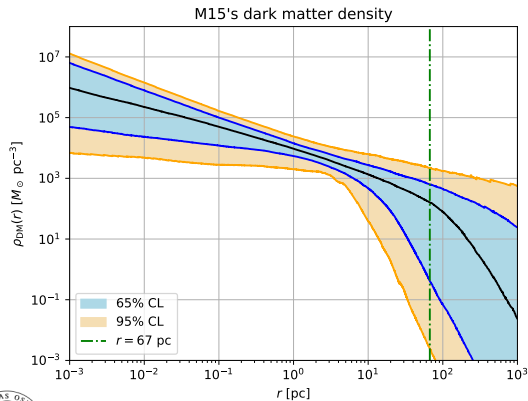


95% of the signal lies in a circle of radius $\sim 0.33^\circ \gg$ MAGIC's resolution

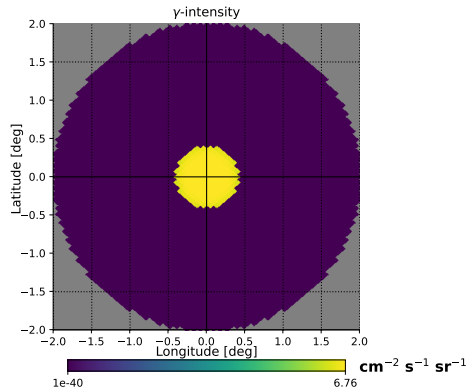
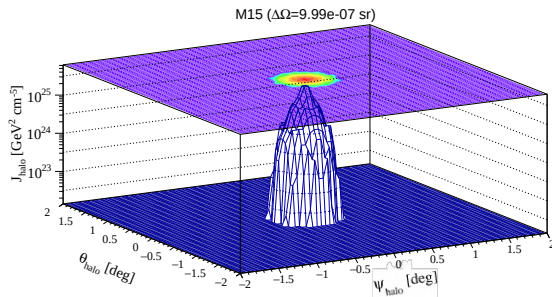
Zhao-Hernquist model (Zhao, 1995; and Hernquist 1990)

$$\rho_{\text{DM}}(r) = \frac{2^{\frac{\beta-\gamma}{\alpha}} \cdot \rho_s}{\left(\frac{r}{r_s}\right)^{\gamma} \cdot \left[1 + \left(\frac{r}{r_s}\right)^{\alpha}\right]^{\frac{\beta-\gamma}{\alpha}}}$$

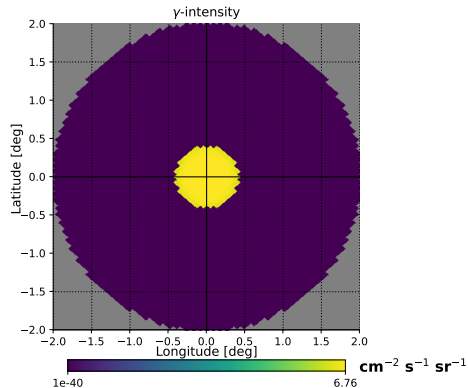
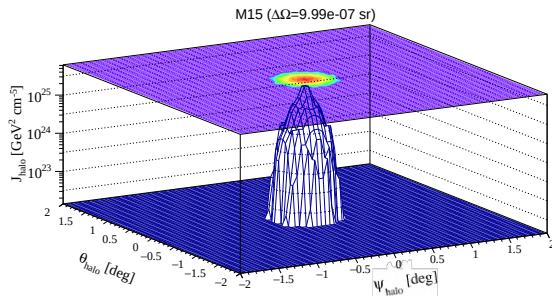
Results



Results



Results



95% of the signal lies in a circle of radius $\sim 0.31^\circ \gg$ MAGIC's resolution



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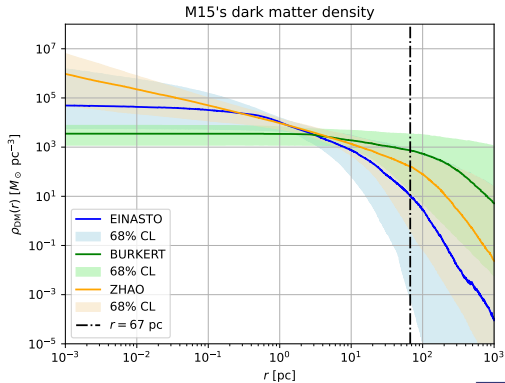
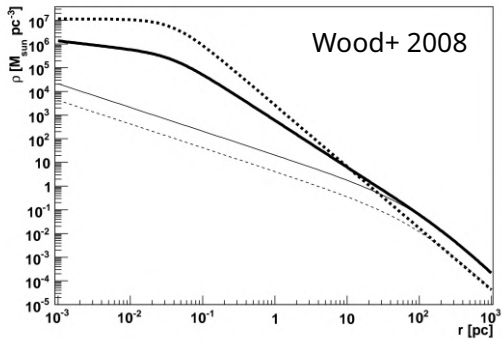


Parameters

Parameter	EINASTO	BURKERT	ZHAO
$\log_{10}(\rho_s/(M_\odot/\text{kpc}^3))$	$11.7^{+0.8}_{-1.1}$	$12.5^{+0.3}_{-0.5}$	$11.7^{+0.9}_{-1.0}$
$\log_{10}(r_s/\text{kpc})$	$-1.7^{+1.6}_{-0.9}$	$-0.8^{+0.4}_{-1.1}$	$-0.9^{+1.2}_{-1.0}$
α	$0.4^{+0.4}_{-0.2}$	-	$1.8^{+0.8}_{-1.3}$
β	-	-	4.9 ± 1.3
γ	-	-	$0.7^{+0.2}_{-0.4}$
$10\beta_0$	$3.4^{+1.9}_{-2.7}$	$5.1^{+1.0}_{-1.3}$	$3.4^{+1.7}_{-2.2}$
β_∞	$-3.7^{+3.4}_{-3.3}$	$-3.45^{+3.1}_{-3.0}$	$-3.7^{+3.2}_{-3.3}$
$\log_{10}(r_a/\text{kpc})$	$0.10^{+2.56}_{-0.08}$	$0.2^{+2.0}_{-0.2}$	$0.13^{+1.97}_{-0.11}$
η	2.2 ± 1.2	$2.3^{+1.1}_{-1.3}$	$2.2^{+1.2}_{-1.3}$
$\log_{10}(J(\alpha_{int} = 0.1^\circ)/(\text{GeV}^2/\text{cm}^5))$	$23.9^{+1.2}_{-0.6}$	$25.0^{+1.0}_{-0.9}$	$24.4^{+0.7}_{-0.8}$
$\log_{10}(J(\alpha_{int} = 0.5^\circ)/(\text{GeV}^2/\text{cm}^5))$	$24.0^{+1.5}_{-0.6}$	$25.7^{+1.2}_{-1.4}$	$24.8^{+0.9}_{-1.0}$
$\log_{10}(J(\alpha_{int} = 10^\circ)/(\text{GeV}^2/\text{cm}^5))$	$24.0^{+1.5}_{-0.6}$	$25.7^{+1.2}_{-1.4}$	$24.8^{+0.9}_{-1.0}$

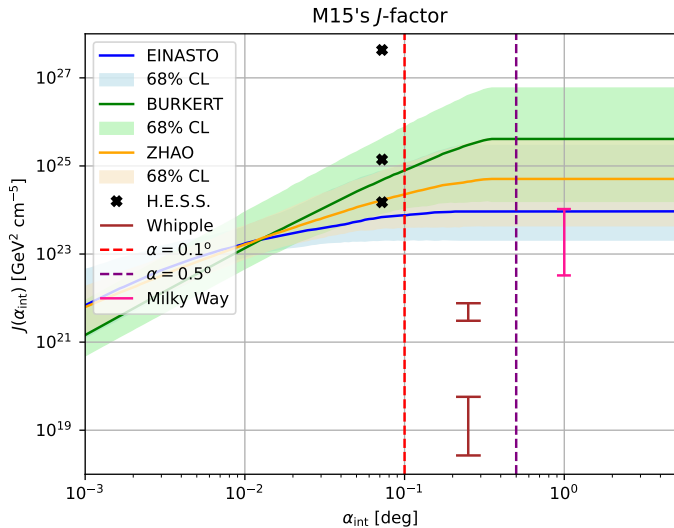


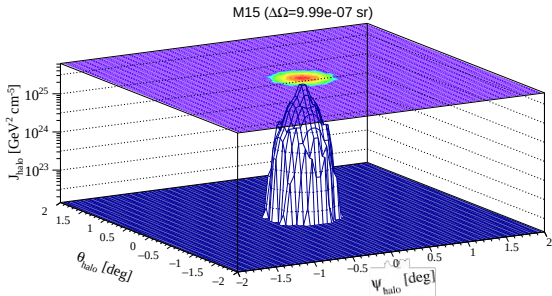
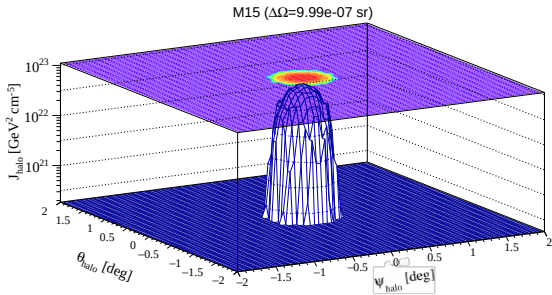
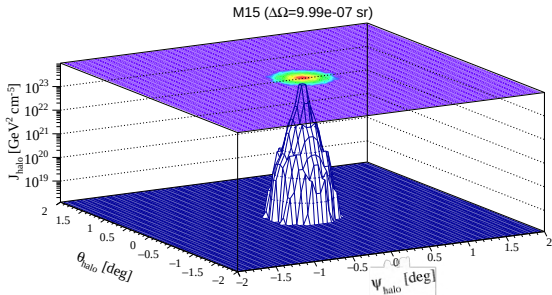
DM density profiles



NFW profile (+AC).

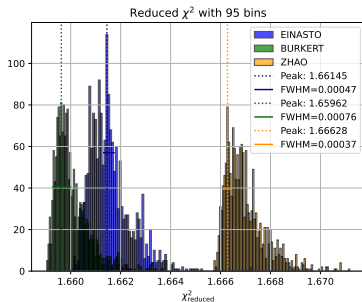
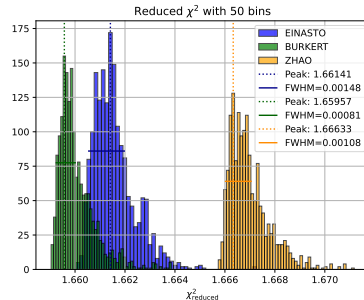
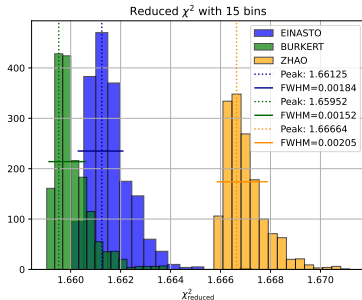
J-factor profile





- Top-left: Einasto model ($r_{95\%} = 0.15^\circ$)
- Top-right: Burkert model ($r_{95\%} = 0.33^\circ$)
- Bottom-left: Zhao-Hernquist model ($r_{95\%} = 0.31^\circ$)



χ^2_{reduced}


Closest peak to 1 : 1.65962 ± 0.00009

Smallest FWHM: 0.00086 ± 00.00022

Best model: **Burkert**

F-test

ZHAO vs EINASTO: F -statistic = 1.09, p -value = $7.40 \cdot 10^{-2}$, $\sigma = 1.79$

ZHAO vs BURKERT: F -statistic = 1.31, p -value = $4.65 \cdot 10^{-8}$, $\sigma = 5.46$

EINASTO vs BURKERT: F -statistic = 1.20, p -value = $1.36 \cdot 10^{-4}$, $\sigma = 3.82$



F-test

ZHAO vs EINASTO: F -statistic = 1.09, p -value = $7.40 \cdot 10^{-2}$, $\sigma = 1.79$

ZHAO vs BURKERT: F -statistic = 1.31, p -value = $4.65 \cdot 10^{-8}$, $\sigma = 5.46$

EINASTO vs BURKERT: F -statistic = 1.20, p -value = $1.36 \cdot 10^{-4}$, $\sigma = 3.82$

Burkert profile is **discarded**.



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Summary

- M15 J -factor: **direct MCMC Jeans analysis** on brightness and kinematic data (**no a priori assumptions on the stellar velocity dispersion**).
- $J(0.5^\circ) \in [2.5 \cdot 10^{23}, 8 \cdot 10^{26}] \text{ GeV}^2 \text{ cm}^{-5}$ (similar to H.E.S.S.).
- Tidal angle falls already in the saturation region.
- The best-fitting model corresponds to a cuspy profile.
- 2/3 models predict M15 being an extended source.
- **Finalizing the DM analysis on the M15 MAGIC data** (publication in preparation).



UNIVERSITY OF OSLO

Analysis of the dark matter density profile of the globular cluster M15

Iván Villegas Pérez
ivanvi@uio.no
Department of Physics

Master Thesis exam

Oslo, June 11th 2025



A. Block/Mt. Lemmon SkyCenter/Univ. of Arizona



Back-up material



2nd order Jeans equation

Liouville equation

$$\frac{\partial f^{(6N)}}{\partial t} + \langle \nabla f^{(6N)}, \mathbf{w}^{(6N)} \rangle = 0$$

$$\mathbf{w} = (\mathbf{v}, -\nabla \phi_{\text{tot}})$$



2nd order Jeans equation

Liouville equation

$$\frac{\partial f^{(6N)}}{\partial t} + \langle \nabla f^{(6N)}, \mathbf{w}^{(6N)} \rangle = 0 \longrightarrow$$

$$\mathbf{w} = (\mathbf{v}, -\nabla \phi_{\text{tot}})$$

Collisionless Boltzmann equation

$$\left\{ \begin{array}{l} \frac{\partial f}{\partial t} + \left\langle \mathbf{v}, \frac{\partial f}{\partial \mathbf{r}} \right\rangle - \left\langle \nabla \phi_{\text{tot}}, \frac{\partial f}{\partial \mathbf{v}} \right\rangle = 0 \\ \phi_{\text{tot}} = \phi + \phi_{\text{ext}} \\ \nabla^2 \phi = -4\pi G \int_{\mathbb{R}^3} f d^3 \mathbf{v} \end{array} \right.$$

2nd order Jeans equation

Liouville equation

$$\frac{\partial f^{(6N)}}{\partial t} + \langle \nabla f^{(6N)}, \mathbf{w}^{(6N)} \rangle = 0 \longrightarrow$$

$$\mathbf{w} = (\mathbf{v}, -\nabla \phi_{\text{tot}})$$

Collisionless Boltzmann equation

$$\left\{ \begin{array}{l} \frac{\partial f}{\partial t} + \left\langle \mathbf{v}, \frac{\partial f}{\partial \mathbf{r}} \right\rangle - \left\langle \nabla \phi_{\text{tot}}, \frac{\partial f}{\partial \mathbf{v}} \right\rangle = 0 \\ \phi_{\text{tot}} = \phi + \phi_{\text{ext}} \\ \nabla^2 \phi = -4\pi G \int_{\mathbb{R}^3} f d^3 \mathbf{v} \end{array} \right.$$

↓

Jeans equation (any symmetry)

$$\left\{ \begin{array}{l} \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} (\rho \bar{v}_i) = 0 \\ \frac{\partial}{\partial t} (\rho \bar{v}_i) + \frac{\partial}{\partial x_j} (\rho v_i \bar{v}_j) + \rho \frac{\partial \phi}{\partial x_i} = 0 \end{array} \right.$$

2nd order Jeans equation

Liouville equation

$$\frac{\partial f^{(6N)}}{\partial t} + \langle \nabla f^{(6N)}, \mathbf{w}^{(6N)} \rangle = 0 \longrightarrow$$

$$\mathbf{w} = (\mathbf{v}, -\nabla \phi_{\text{tot}})$$

Spherical Jeans equation

$$\frac{1}{n_*} \left[\frac{d}{dr} (n_* \bar{v}_r^2) \right] + \frac{2}{r} \beta_{\text{ani}}(r) \bar{v}_r^2 =$$

$$= -\frac{G[M_{\text{DM}}(r) + M_*(r)]}{r^2}$$

Collisionless Boltzmann equation

$$\left\{ \begin{array}{l} \frac{\partial f}{\partial t} + \left\langle \mathbf{v}, \frac{\partial f}{\partial \mathbf{r}} \right\rangle - \left\langle \nabla \phi_{\text{tot}}, \frac{\partial f}{\partial \mathbf{v}} \right\rangle = 0 \\ \phi_{\text{tot}} = \phi + \phi_{\text{ext}} \\ \nabla^2 \phi = -4\pi G \int_{\mathbb{R}^3} f d^3 \mathbf{v} \end{array} \right.$$

↓
← Jeans equation (any symmetry)

$$\left\{ \begin{array}{l} \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i} (\rho \bar{v}_i) = 0 \\ \frac{\partial}{\partial t} (\rho \bar{v}_i) + \frac{\partial}{\partial x_j} (\rho \bar{v}_i \bar{v}_j) + \rho \frac{\partial \phi}{\partial x_i} = 0 \end{array} \right.$$

Einasto model (Einasto, 1965)

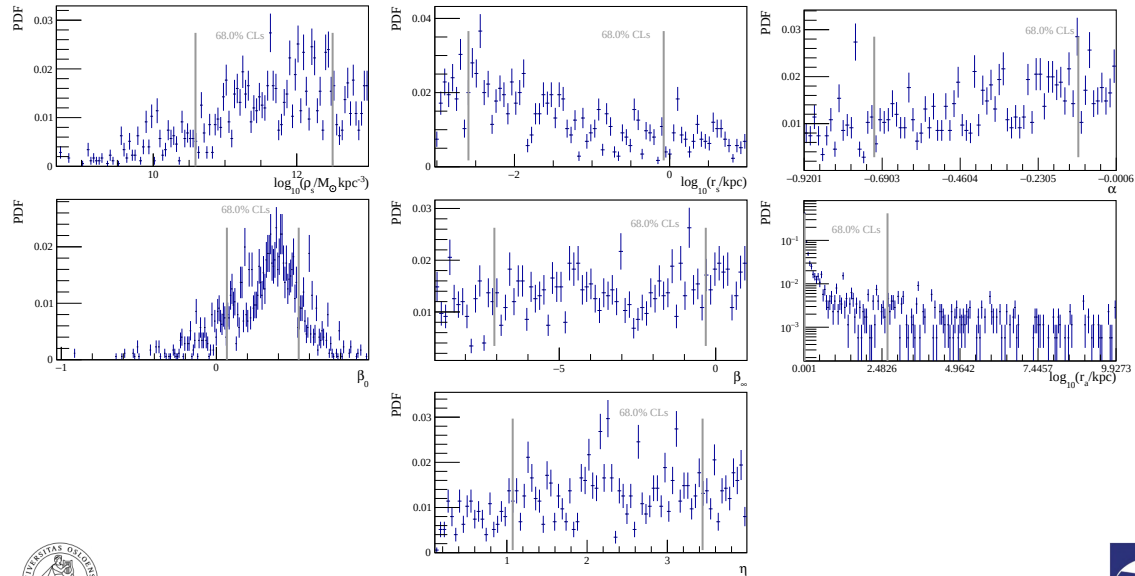
$$\rho_{\text{DM}}(r) = \rho_s \cdot e^{-\frac{2}{\alpha} \left[\left(\frac{r}{r_s} \right)^\alpha - 1 \right]}$$

$$R_t = 44 \text{ pc}$$

Parameter	Median	68% CL _{low}	68% CL _{up}
$\log_{10}(\rho_s/(M_\odot/\text{kpc}^3))$	11.7	10.6	12.5
$\log_{10}(r_s/\text{kpc})$	-1.7	-2.6	-0.1
α	0.4	0.2	0.8
$10\beta_0$	3.4	0.7	5.3
β_∞	-3.7	-7.0	-3.2
$\log_{10}(r_a/\text{kpc})$	0.10	0.02	2.66
η	2.3	1.0	3.4
$\log_{10}(J(\alpha_{int} = 0.1^\circ)/(\text{GeV}^2/\text{cm}^5))$	23.9	23.3	25.0
$\log_{10}(J(\alpha_{int} = 0.5^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.0	23.3	25.5
$\log_{10}(J(\alpha_{int} = 10^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.0	23.3	25.5
$\log_{10}(J(\alpha_{int} = \alpha_t)/(\text{GeV}^2/\text{cm}^5))$	24.0	23.3	25.5

$$\alpha_t = R_t/d, \text{ where } d = 10.94 \text{ kpc.}$$



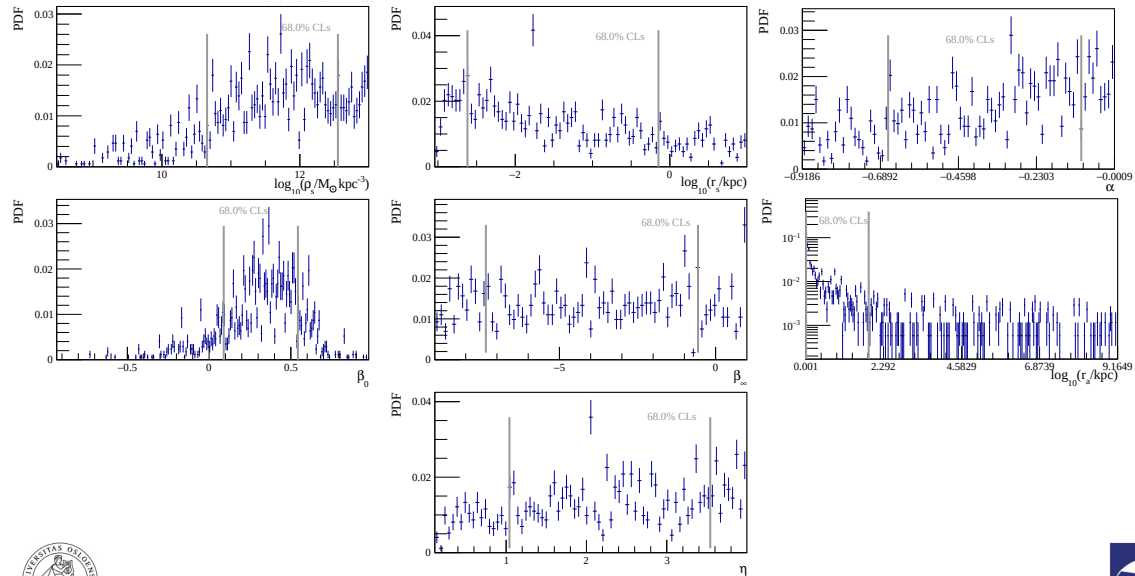


$$R_t = 58 \text{ pc}$$

Parameter	Median	68% CL _{low}	68% CL _{up}
$\log_{10}(\rho_s/(M_\odot/\text{kpc}^3))$	11.7	10.7	12.6
$\log_{10}(r_s/\text{kpc})$	-1.70	-2.62	-0.15
α	0.5	0.2	0.8
$10\beta_0$	3.5	0.9	5.4
β_∞	-3.9	-7.4	-0.6
$\log_{10}(r_a/\text{kpc})$	0.10	0.02	1.86
η	2.3	1.0	3.5
$\log_{10}(J(\alpha_{int} = 0.1^\circ)/(\text{GeV}^2/\text{cm}^5))$	23.9	23.3	25.1
$\log_{10}(J(\alpha_{int} = 0.5^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.0	23.3	25.5
$\log_{10}(J(\alpha_{int} = 10^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.0	23.3	25.5
$\log_{10}(J(\alpha_{int} = \alpha_t)/(\text{GeV}^2/\text{cm}^5))$	24.1	23.3	25.7

$$\alpha_t = R_t/d, \text{ where } d = 10.94 \text{ kpc.}$$



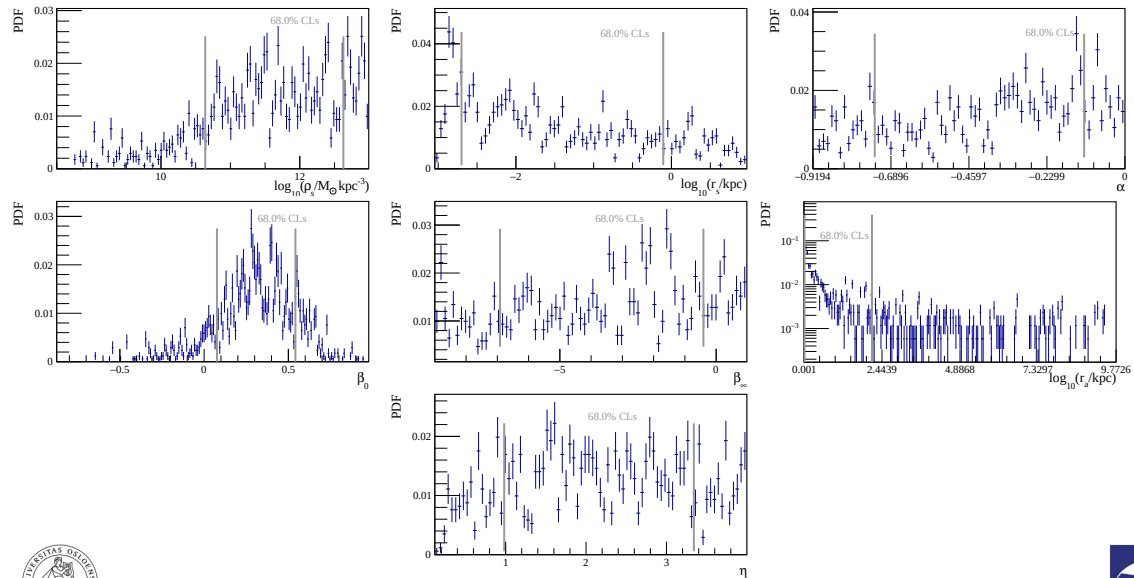


$$R_t = 67 \text{ pc}$$

Parameter	Median	68% CL_{low}	68% CL_{up}
$\log_{10}(\rho_s/(M_{\odot}/\text{kpc}^3))$	11.7	10.6	12.6
$\log_{10}(r_s/\text{kpc})$	-1.7	-2.7	-0.1
α	0.4	0.2	0.8
β_{∞}	-3.3	-6.9	-4.1
$\log_{10}(r_a/\text{kpc})$	0.12	0.02	2.13
η	2.1	1.0	3.3
$\log_{10}(J(\alpha_{\text{int}} = 0.1^{\circ})/(\text{GeV}^2/\text{cm}^5))$	23.9	23.4	25.1
$\log_{10}(J(\alpha_{\text{int}} = 0.5^{\circ})/(\text{GeV}^2/\text{cm}^5))$	23.9	23.3	25.8
$\log_{10}(J(\alpha_{\text{int}} = 10^{\circ})/(\text{GeV}^2/\text{cm}^5))$	23.9	23.3	25.8
$\log_{10}(J(\alpha_{\text{int}} = \alpha_t)/(\text{GeV}^2/\text{cm}^5))$	23.9	23.3	25.8

$$\alpha_t = R_t/d, \text{ where } d = 10.94 \text{ kpc.}$$



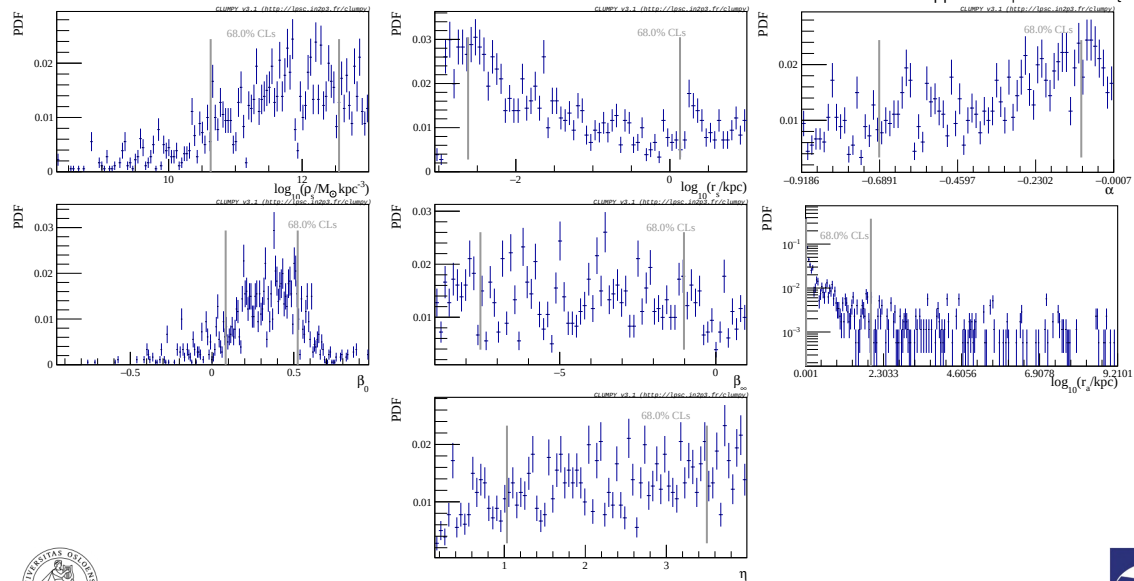


$$R_t = 80 \text{ pc}$$

Parameter	Median	68% CL _{low}	68% CL _{up}
$\log_{10}(\rho_s/(M_\odot/\text{kpc}^3))$	11.6	10.7	12.6
$\log_{10}(r_s/\text{kpc})$	-1.38	-2.65	0.03
α	0.5	0.2	0.8
$10\beta_0$	3.4	0.9	5.4
β_∞	-3.9	-6.9	-0.6
$\log_{10}(r_a/\text{kpc})$	0.11	0.02	1.93
η	2.2	1.0	3.4
$\log_{10}(J(\alpha_{int} = 0.1^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.1	23.4	25.6
$\log_{10}(J(\alpha_{int} = 0.5^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.3	23.4	26.3
$\log_{10}(J(\alpha_{int} = 10^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.4	23.4	26.3
$\log_{10}(J(\alpha_{int} = \alpha_t)/(\text{GeV}^2/\text{cm}^5))$	24.3	23.4	26.3

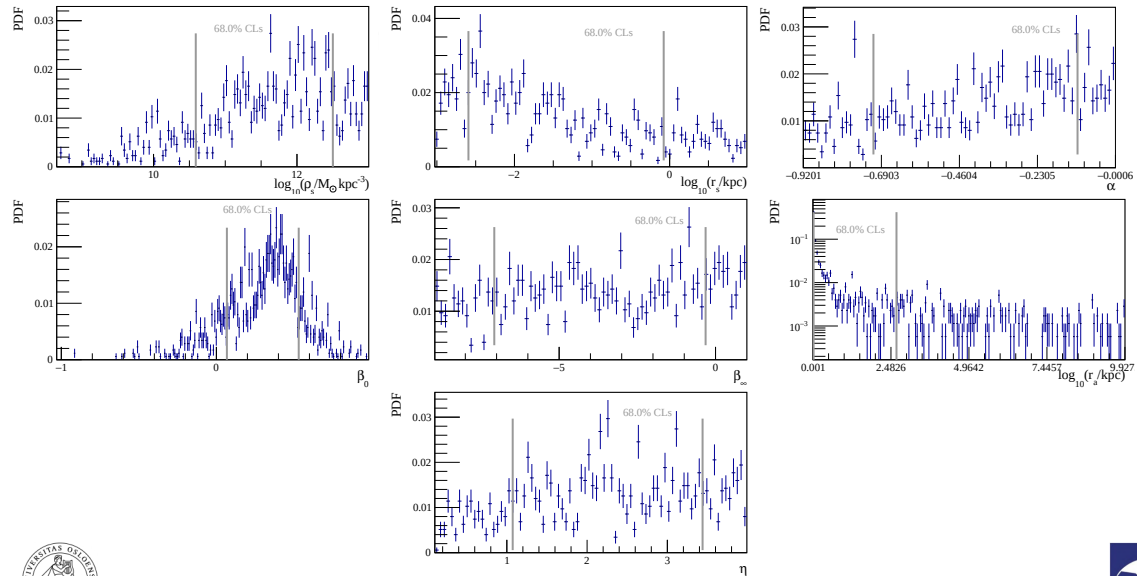
$$\alpha_t = R_t/d, \text{ where } d = 10.94 \text{ kpc.}$$





All simulations combined

Parameter	Median	68% CL _{low}	68% CL _{up}
$\log_{10}(\rho_s/(M_\odot/\text{kpc}^3))$	11.7	10.6	12.5
$\log_{10}(r_s/\text{kpc})$	-1.7	-2.6	-0.1
α	0.4	0.2	0.8
$10\beta_0$	3.4	0.7	5.3
β_∞	-3.7	-7.1	-0.3
$\log_{10}(r_a/\text{kpc})$	0.10	0.02	2.66
η	2.3	1.1	3.4
$\log_{10}(J(\alpha_{int} = 0.1^\circ)/(\text{GeV}^2/\text{cm}^5))$	23.9	23.4	25.1
$\log_{10}(J(\alpha_{int} = 0.5^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.0	23.4	25.5
$\log_{10}(J(\alpha_{int} = 10^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.0	23.4	25.5

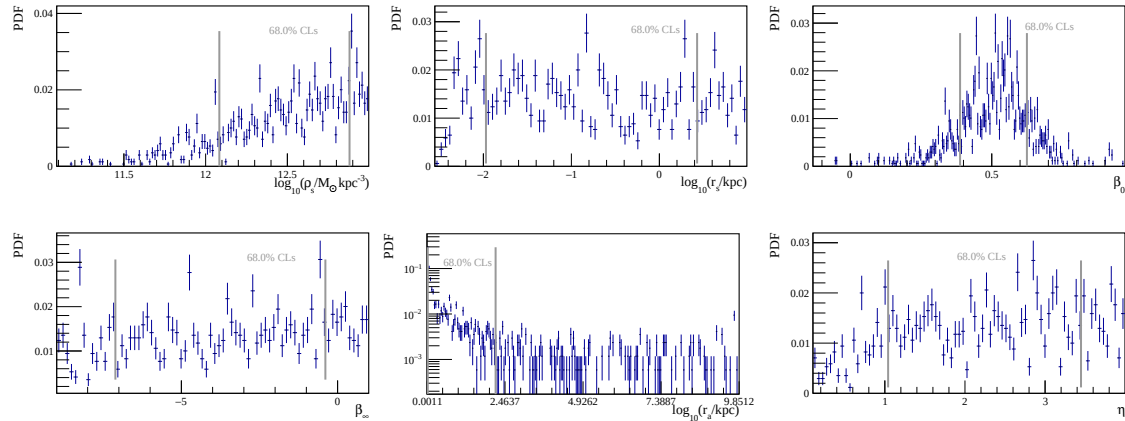


Burkert model (Burkert, 1995)

$$\rho_{\text{DM}}(r) = \frac{\rho_s}{\left(1 + \frac{r}{r_s}\right) \cdot \left[1 + \left(\frac{r}{r_s}\right)^2\right]}$$

$$R_t = 67 \text{ pc}$$

Parameter	Median	68% CL _{low}	68% CL _{up}
$\log_{10}(\rho_s/(M_\odot/\text{kpc}^3))$	12.5	12.1	12.9
$\log_{10}(r_s/\text{kpc})$	-0.9	-2.0	-0.4
$10\beta_0$	5.1	3.9	6.2
β_∞	-3.5	-7.1	-0.4
$\log_{10}(r_a/\text{kpc})$	0.18	0.02	2.17
η	2.3	1.0	3.4
$\log_{10}(J(\alpha_{int} = 0.1^\circ)/(\text{GeV}^2/\text{cm}^5))$	25.0	24.1	25.9
$\log_{10}(J(\alpha_{int} = 0.5^\circ)/(\text{GeV}^2/\text{cm}^5))$	25.7	24.2	26.8
$\log_{10}(J(\alpha_{int} = 10^\circ)/(\text{GeV}^2/\text{cm}^5))$	25.7	24.2	26.8



Zhao-Hernquist model (Zhao, 1995; and Hernquist 1990)

$$\rho_{\text{DM}}(r) = \frac{2^{\frac{\beta-\gamma}{\alpha}} \cdot \rho_s}{\left(\frac{r}{r_s}\right)^{\gamma} \cdot \left[1 + \left(\frac{r}{r_s}\right)^{\alpha}\right]^{\frac{\beta-\gamma}{\alpha}}}$$

$$R_t = 67 \text{ pc}$$

Parameter	Median	68% CL _{low}	68% CL _{up}
$\log_{10}(\rho_s/(M_\odot/\text{kpc}^3))$	11.7	10.7	12.6
$\log_{10}(r_s/\text{kpc})$	-0.9	-1.9	0.3
α	1.8	0.5	2.6
β	4.9	3.6	6.2
γ	0.7	0.3	0.9
$10\beta_0$	3.4	1.2	5.1
β_∞	-3.7	-7.0	-0.5
$\log_{10}(r_a/\text{kpc})$	0.13	0.02	2.11
η	2.2	1.0	3.4
$\log_{10}(J(\alpha_{int} = 0.1^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.4	23.7	25.1
$\log_{10}(J(\alpha_{int} = 0.5^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.8	23.7	25.7
$\log_{10}(J(\alpha_{int} = 10^\circ)/(\text{GeV}^2/\text{cm}^5))$	24.8	23.7	25.7



